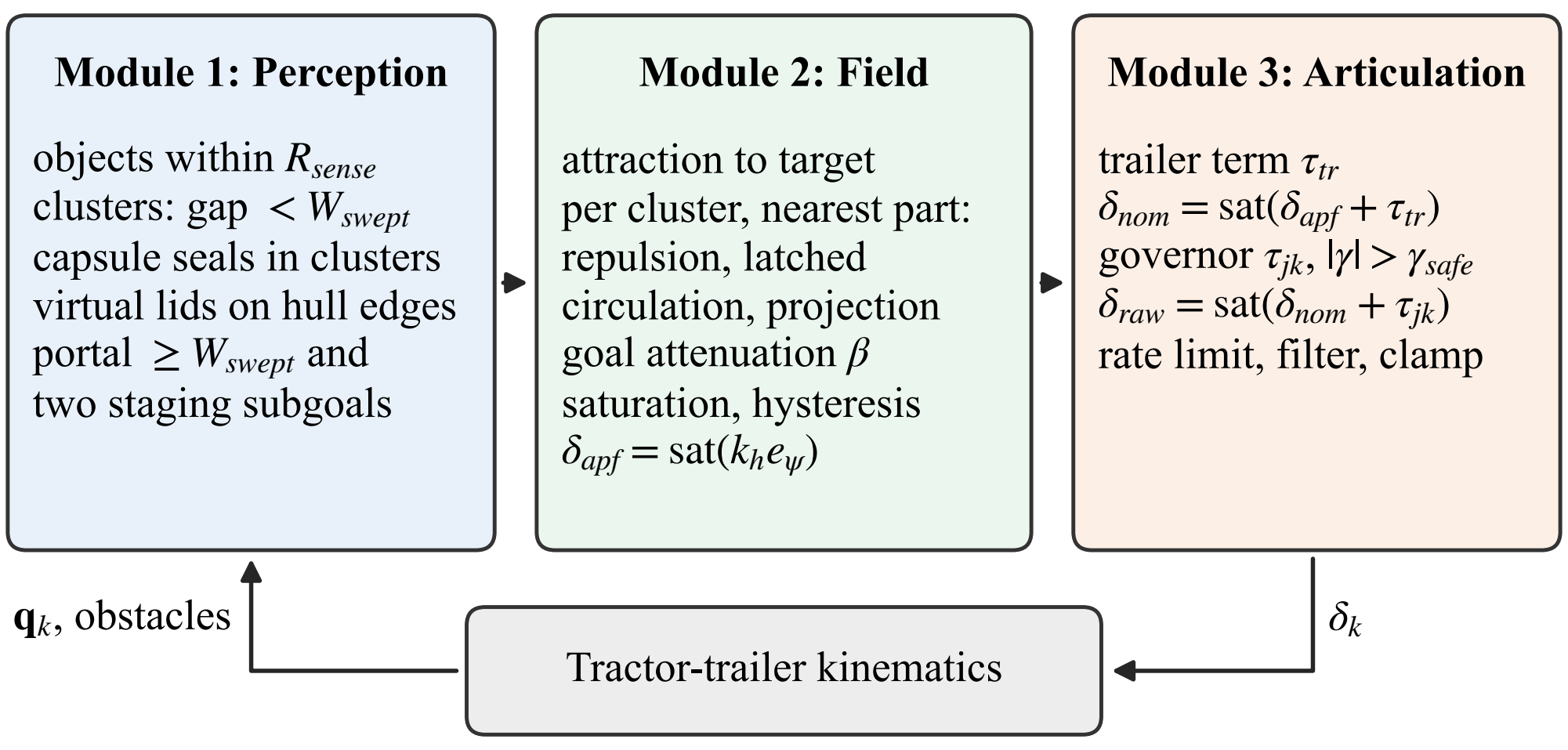

Module 1: Perception
objects within R_{sense}
clusters: gap $< W_{swept}$
capsule seals in clusters
virtual lids on hull edges
portal $\geq W_{swept}$ and
two staging subgoals
Module 2: Field
attraction to target
per cluster, nearest part:
repulsion, latched
circulation, projection
goal attenuation β
saturation, hysteresis
$\delta_{apf} = \mathrm{sat}(k_h e_\psi)$
Module 3: Articulation
trailer term τ_{tr}
$\delta_{nom} = \mathrm{sat}(\delta_{apf} + \tau_{tr})$
governor τ_{jk}, $|\gamma| > \gamma_{safe}$
$\delta_{raw} = \mathrm{sat}(\delta_{nom} + \tau_{jk})$
rate limit, filter, clamp
$\mathbf{q}_k$, obstacles
Tractor-trailer kinematics
δ_k